1. Run report: EXP-ECDLP-20a588 — m=3 block order, L=3..7

Observations only. This report records what ran and what was measured. It has no verdict, support or weaken language; those belong to the evidence review.

- **Experiment** EXP-ECDLP-20a588 · **hypothesis** H-ECDLP-60de36 · **question** RQ-ECDLP-001 · **goal** GOAL-ECDLP-001 · **protocol** `EXP-ECDLP-20a588/specification.yaml` with its amendments.
- **Host** 4 vCPU Intel Xeon 2.10 GHz, 15 GB; msolve 0.6.5-1build2 (1 thread per process); Singular 4.3.2. Four worker processes, one solver process per solve. Timeout 600 s per solve. Seeds: sha256 of `EXP-ECDLP-f49f75/v1|m|L|instance`.

1.1. Runs

run	started (UTC)	finished	rows	git HEAD
20a588-01	2026-10-15 23:06	2026-10-16 01:07	24	b19cbdaf66
20a588-02	2026-10-19 16:08	none (lost)	0	d2d3f9ef29
20a588-03	2026-10-19 16:18	2026-10-19 16:58	6	1b1f8c2d02

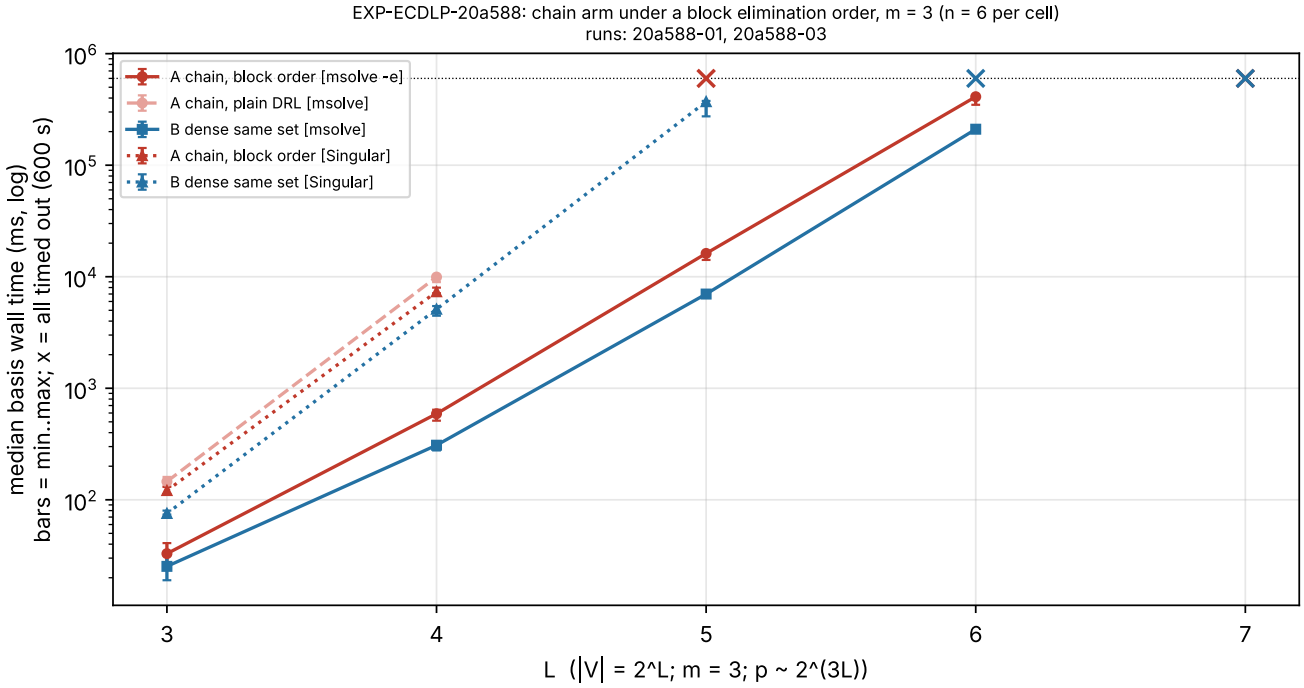


Figure 1: Median basis time per arm vs L (measured; process wall time), bars min–max over completed instances; × = every instance timed out (600 s). Source runs as in the table. Where a cell mixes two primes (EXP-ECDLP-8c41d7, L = 4: Singular arms at p = 65551, msolve arms at p = 262271) only comparisons within one solver family are on shared instances.

1.2. Per-cell measurements

Columns: completed / timed out / other failure / degenerate target; median ms [min–max]; planted-in-variety (passed/checked); max F4 degree (msolve arms). A cell can mix two runs (and two primes, `p bits`) where an amendment re-ran arms; the source runs are listed.

L	p bits	arm	runs	ok/to/oth/deg	median ms [min–max]	planted	F4 deg
3	10	A_block_sg	01	6/0/0/0	121 [112–130]	4/4	—
3	10	A_drl_ms	01	6/0/0/0	146 [133–160]	4/4	12
3	10	A_elim_ms	01	6/0/0/0	33 [27–41]	4/4	14
3	10	B_ms	01	6/0/0/0	25 [19–32]	4/4	16
3	10	B_sg	01	6/0/0/0	75 [74–80]	4/4	—
4	13	A_block_sg	01	6/0/0/0	7,329 [7,116–7,984]	4/4	—
4	13	A_drl_ms	01	6/0/0/0	9,933 [8,947–10,140]	4/4	13
4	13	A_elim_ms	01	6/0/0/0	591 [512–642]	4/4	32
4	13	B_ms	01	6/0/0/0	309 [277–336]	4/4	24

4	13	B_sg	01	6/0/0/0	5,112 [4,472–5,458]	4/4	—
5	16	A_block_sg	01	0/6/0/0	— [——]	—	—
5	16	A_drl_ms	01	0/6/0/0	— [——]	—	—
5	16	A_elim_ms	01	6/0/0/0	16,183 [14,145–16,526]	4/4	65
5	16	B_ms	01	6/0/0/0	6,993 [6,559–7,195]	4/4	40
5	16	B_sg	01	6/0/0/0	370,938 [274,132–376,859]	4/4	—
6	19	A_elim_ms	01	6/0/0/0	411,159 [347,985–418,707]	4/4	133
6	19	B_ms	01	6/0/0/0	210,271 [194,599–213,522]	4/4	72
6	19	B_sg	01	0/6/0/0	— [——]	—	—
7	22	A_elim_ms	03	0/6/0/0	— [——]	—	—
7	22	B_ms	03	0/6/0/0	— [——]	—	—

1.3. Operational notes (facts, not findings)

- RUN-01 was stopped by the host's 7200 s background-task limit during L = 7; cells L = 3..6 were complete. RUN-02 (the AMD-01 run for L = 7) was lost to a VM recycle after about ten minutes and wrote no rows; RUN-03 (AMD-02) re-ran L = 7 for the two arms the stopping rule had not retired (A_elim_ms, B_ms).
- Stopping rule: A_drl_ms and A_block_sg were retired at L = 5, B_sg at L = 6; at L = 7 every instance timed out for both A_elim_ms and B_ms.
- Cell L = 6: A_elim_ms 411 s and B_ms 210 s median (completed); the Singular dense arm timed out on all six instances.
- msolve 0.6.5 segfaults on systems over 17- and 18-bit primes (CORR-20261019-e6f65b); no cell of this experiment used one (primes 10, 13, 16, 19 and 22 bits).
- A unit-test run (tools/test_pkm_isogeny_launchers.py, about 6 s on one core) and a few seconds of exploratory analysis shared the host with RUN-02/RUN-03 (L = 7, where every solve ended at the 600 s cap, so the timings are not affected).
- Every planted decomposition found lies in the computed basis; the eliminated chain basis equals the dense basis on every instance where both finished.